

Türkisch-Deutsche Universität

ETE 412

Biomedical Signal Processing

Laboratory Project Report

Design and Implementation of a High-Precision Signal Conditioning Circuit for Electromyography (EMG) Applications

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Abstract

This report details the architectural design, mathematical modeling, simulation, and hardware implementation of a high-precision, low-noise signal conditioning circuit for surface Electromyography (sEMG) acquisition. Electromyographic signals extracted from skeletal muscles exhibit extremely low amplitudes ($1\,\mu\text{V} - 10\,\text{mV}$) and are heavily corrupted by ambient electromagnetic interference, specifically the 50 Hz power line hum, as well as high-frequency thermal noise and motion artifacts. To extract clean bio-potential waveforms suitable for analog-to-digital conversion (ADC) via an ESP32 microcontroller, a multi-stage analog processing chain was developed. The system incorporates an instrumentation amplifier (INA121) for high Common-Mode Rejection Ratio (CMRR), an active second-order Sallen-Key High-Pass Filter (HPF) for motion artifact suppression, an active Twin-T Notch Filter with adjustable feedback for sharp power-line attenuation, a second-order Sallen-Key Low-Pass Filter (LPF) for anti-aliasing, a final post-gain non-inverting amplifier, and a precision half-wave rectifier coupled with an envelope detector. Circuit parameters were verified using LTspice AC analysis, and the physical substrate was engineered using KiCad for PCB layout and 3D modeling.

1 Introduction and Historical Background

Surface Electromyography (sEMG) is a non-invasive technique utilized to record the electrical activity generated by skeletal muscles during contraction. The acquisition of sEMG signals serves as a foundational component in biomedical engineering, prosthetics control, rehabilitation robotics, and human-machine interfaces.

1.1 Chronological Evolution of Electromyography

The scientific foundation of electromyography evolved over multiple centuries, transitioning from early biological observations to advanced microelectronic processing architectures:

- **The Inception of Bioelectricity (18th Century):** The historical lineage of EMG dates back to the pioneering experiments of Luigi Galvani in 1791, who demonstrated that electrical stimulation could induce muscular contractions in frog limbs. This critical milestone established the fundamental concept of bioelectricity, proving that muscular movement is inherently tied to electrical impulses.
- **Detection of Voluntary Potentials (19th Century):** In 1849, Carlo Matteucci detected primitive electrical currents directly originating from contracting active muscle tissue. This was expanded by Emil du Bois-Reymond, who successfully recorded voluntary bioelectric activity in human skeletal muscle loops, proving that electrical discharge occurs concurrently with muscle contractions under conscious control.
- **The String Galvanometer Era (Early 20th Century):** The detection accuracy of biological signals drastically improved in the early 1900s through Willem Einthoven's development of the string galvanometer. In 1928, E.D. Adrian and D.W. Bronk revolutionized the discipline by introducing the coaxial needle electrode, allowing researchers to isolate individual, highly localized Motor Unit Action Potential (MUAP) profiles for the first time.
- **The Solid-State Revolution (Late 20th Century to Present):** Prior to the mid-20th century, electromyography systems were severely limited by bulky vacuum-tube amplification and low-bandwidth analog recording charts. The development of silicon transistors and integrated operational amplifiers in the 1960s allowed for the production of compact, high-precision instrumentation circuits.

1.2 Physiological Mechanism and Signal Integrity Challenges

Skeletal muscles are orchestrated by Motor Units (MUs), which comprise a single motor neuron and the specific group of muscle fibers it innervates. Upon activation by the central nervous system, an action potential propagates down the motor neuron, triggering cellular depolarization across the postsynaptic muscle membranes, yielding a transient extracellular potential distribution known as the Motor Unit Action Potential (MUAP).

Because a functional muscle contraction requires the simultaneous, asynchronous recruitment of thousands of MUAPs, the cumulative electrical field propagates through the conductive internal soft tissues and dermal layers. Surface electromyography utilizes specialized skin-surface electrodes to intercept these compound electrical fluctuations. However, the raw bio-potential signal captured via these surface electrodes presents significant signal integrity challenges:

1. **Extremely Low Amplitude:** Muscle potentials vary between $1\ \mu\text{V}$ and $10\ \text{mV}$ peak-to-peak, requiring substantial voltage amplification before digitization.

2. **Spectral Overlap with Noise:** The dominant energy spectrum of an sEMG signal resides between 20 Hz and 500 Hz. This frequency band is highly susceptible to 50 Hz power line interference, baseline wander caused by motion artifacts (< 20 Hz), and high-frequency thermal or instrumentation noise (> 500 Hz).

1.3 Proposed System Architecture and Functional Blocks

To overcome these severe physiological and environmental constraints, this project focuses on designing a robust analog signal conditioning frontend. The signal processing path is systematically divided into distinct operational blocks to transform the raw microvolt-level differential signal into a clean DC metric:

- **Instrumentation Amplification:** Provides a high common-mode rejection baseline to suppress common environmental noise.
- **Active High-Pass Filtering:** Eliminates low-frequency motion artifacts and baseline wander below 19.41 Hz.
- **Active Twin-T Notch Rejection:** Enforces a deep, localized attenuation drop precisely at the disruptive 50.05 Hz power line node.
- **Active Low-Pass Filtering:** Serves as an anti-aliasing filter by restricting high-frequency thermal and instrumentation noise above 482.28 Hz.
- **Active Precision Rectifier and Envelope Tracking:** Overrides standard diode forward voltage drops to linearize low-amplitude variations and extracts a smooth single-ended DC control envelope.

The final output is tailored specifically to comply with the $0\text{ V} - 3.3\text{ V}$ dynamic input range of an external microcontroller’s Analog-to-Digital Converter (ADC), preparing the biological signal for safe and precise downstream data acquisition.

2 Mathematical Modeling and Circuit Topology Justification

Each stage of the signal conditioning circuit is mathematically optimized to guarantee signal fidelity, stability, and ideal frequency response.

2.1 Stage 1: Front-End Instrumentation Amplification

The primary abstraction layer of the analog frontend utilizes a high-performance instrumentation amplifier (In-Amp) topology. For this implementation, the INA121 precision instrumentation amplifier was specifically selected over conventional bipolar alternatives due to its specialized architectural characteristics tailored for biomedical potential acquisition:

1. **Ultra-High Input Impedance and FET Inputs:** The INA121 features Field-Effect Transistor (FET) inputs, providing an exceptionally high input impedance of $10^{12}\ \Omega \parallel 3\text{ pF}$. Because surface EMG signals are captured via skin-electrode interfaces that exhibit inherently high and varying source impedances ($1\text{ k}\Omega$ to $> 100\text{ k}\Omega$), a massive input impedance is mathematically mandatory to minimize loading effects and signal attenuation according to the voltage divider rule.
2. **Extremely Low Input Bias Current:** Bipolar instrumentation amplifiers often suffer from continuous input bias currents that polarize the electrodes, leading to severe DC baseline drift and half-cell potential saturation. The INA121 restricts this parameter to a maximum of 4 pA (50 pA absolute maximum at elevated temperatures), completely preserving the electrochemical stability of the skin-electrode interface.

3. **Superior Common-Mode Rejection Ratio (CMRR):** Bio-potentials exhibit substantial common-mode voltage noise injected into the human body from surrounding 50 Hz electromagnetic fields. The INA121 delivers a CMRR exceeding 106 dB at low structural gains, ensuring that common-mode noise common to both sensing nodes is strongly suppressed before the differential muscle potential undergoes primary amplification.

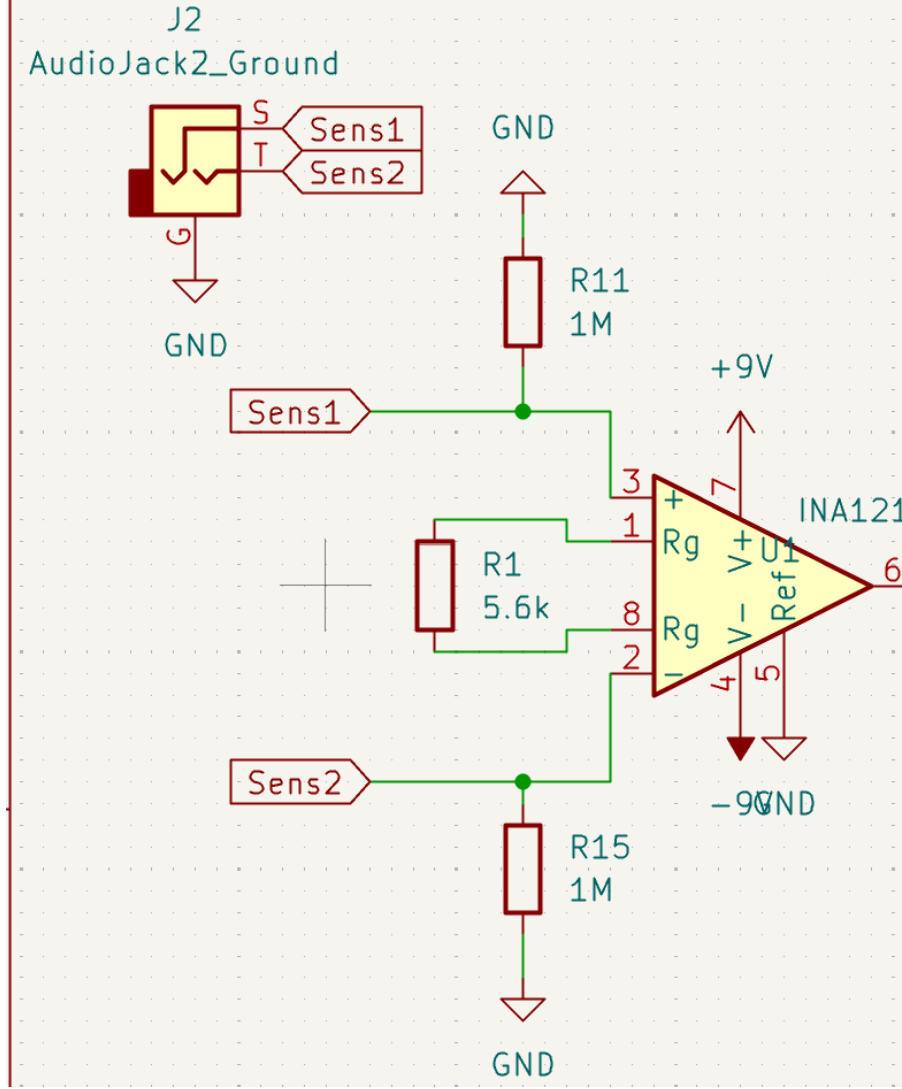


Figure 1: Internal schematic and input bias current return path configuration of the INA121 precision instrumentation amplifier with external passive components.

The overall differential gain A_d is established via a single external resistor R_G across pins 1 and 8, as structurally mapped in Figure 1, satisfying the transfer equation:

$$A_d = 1 + \frac{50, \text{k}\Omega}{R_G} \quad (1)$$

In this design, a standard $5.6 \text{ k}\Omega$ ($\pm 1\%$, E24/E96 series) metal-film resistor was selected for R_G . Substituting this specific resistance value into the transfer equation yields a precise primary stage differential gain of:

$$A_d = 1 + \frac{50, \text{k}\Omega}{5.6, \text{k}\Omega} \approx 9.93 \quad (2)$$

This targeted gain of approximately 10 provides sufficient initial amplification to raise the microvolt-level surface EMG signals safely above the system noise floor, while maintaining

a crucial linear headroom buffer to avoid saturation from high-amplitude electrode half-cell potential offsets or motion artifacts before reaching the subsequent high-pass filtering stages. The reference pin (V_{ref}) is securely tied to the analog signal ground to maintain a bipolar swing around the reference node, outputting the preliminary voltage profile:

$$V_{out1} = A_d \cdot (V_{in+} - V_{in-}) \quad (3)$$

To provide a continuous return path for the minute input bias currents of the FET gates, two $1\text{ M}\Omega$ ($\pm 1\%$) metal-film resistors (R_{B1} and R_{B2}) are connected symmetrically between the differential inputs (V_{in+} and V_{in-}) and the analog ground. This explicit configuration prevents input stage saturation due to open-circuit floating potentials while successfully keeping the total differential input impedance well within the high-performance threshold required for biopotential sensing

2.2 Stage 2: Second-Order Active High-Pass Filter (HPF)

Motion artifacts, such as electrode displacement and skin-electrode impedance shifts, introduce low-frequency baseline drift. A second-order active Sallen-Key High-Pass Filter configuration is selected due to its non-inverting nature, excellent isolation, and ease of tuning.

In this design, an equal-component topology is implemented where the passive tracking elements are unified such that:

$$R_2 = R_3 = R = 82\text{ k}\Omega, \quad C_1 = C_2 = C = 100\text{ nF} \quad (4)$$

The generic transfer function $H_{HP}(s)$ of a second-order high-pass Sallen-Key filter is defined as:

$$H_{HP}(s) = \frac{A_0 \cdot s^2}{s^2 + \frac{\omega_c}{Q}s + \omega_c^2} \quad (5)$$

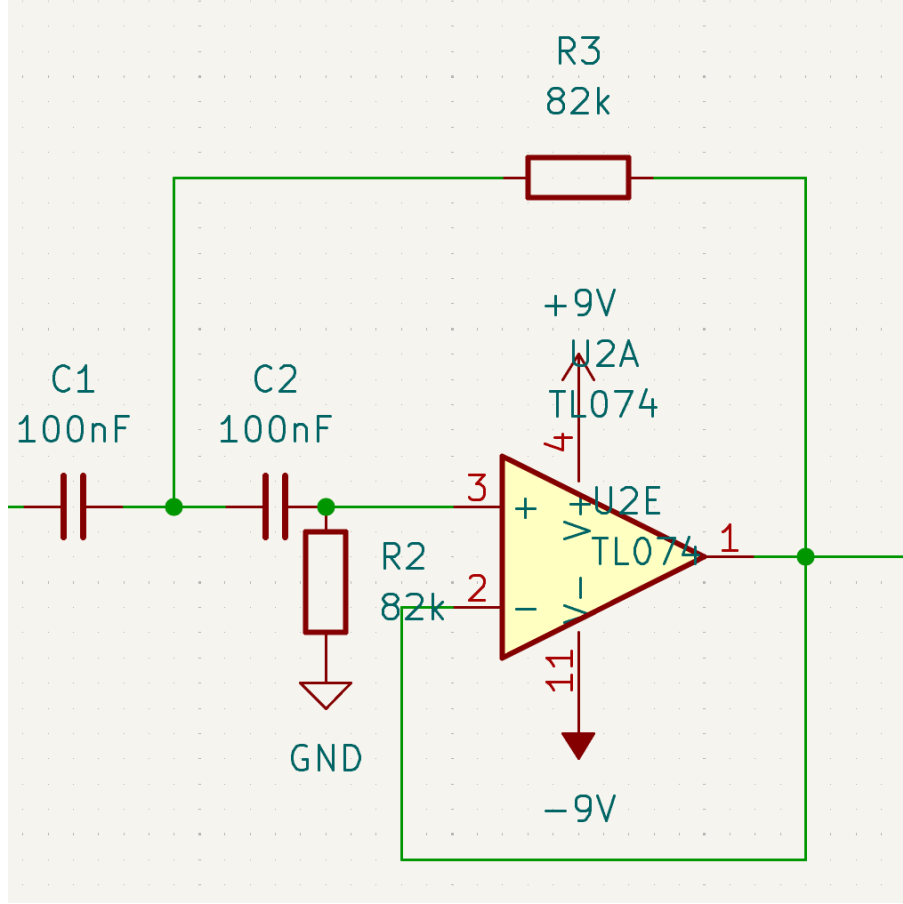


Figure 2: Active second-order Sallen-Key High-Pass Filter configured in an equal-component unity-gain topology.

Where the angular cutoff frequency $\omega_c = 2\pi f_c$ for the equal-component configuration mapped in Figure 2 simplifies precisely to:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 82 \times 10^3 \cdot 100 \times 10^{-9}} \approx 19.41 \text{ Hz} \quad (6)$$

This mathematically establishes the lower cutoff boundary at approximately 19.4 Hz, which safely rejects DC offsets and low-frequency baseline wander without compromising the meaningful lower spectrum of the sEMG signal.

Crucially, because a unity-gain voltage follower setup ($A_0 = 1$) is leveraged with identical component values ($R_1 = R_2$ and $C_1 = C_2$), the quality factor Q of this specific equal-component network evaluates strictly to:

$$Q = \frac{1}{3 - A_0} = \frac{1}{3 - 1} = 0.5 \quad (7)$$

A quality factor of $Q = 0.5$ designates an over-damped, multi-pole exponential decay characteristic. While this introduces a slightly softer transition ("knee") at the cutoff boundary compared to a maximally flat Butterworth response ($Q = 0.707$), it guarantees absolute phase stability and completely eliminates any potential frequency-domain peaking or ringing artifacts near the 20 Hz passband threshold, ensuring an uncorrupted transient response for subsequent filtration stages.

2.3 Stage 3: Active Twin-T Notch Filter (50 Hz Suppression)

A passive Twin-T network introduces deep attenuation features at its center frequency but suffers from an inherently wide rejection bandwidth ($Q = 0.25$), which would inadvertently

clip or distort valid adjacent muscle potentials. To sharpen the transition band and isolate the power-line interference node, an active bootstrap feedback loop is introduced via an operational amplifier buffer configuration.

The active Twin-T branch topologies are constructed with a high degree of mathematical balance based on the tracking component configurations:

$$R_{13} = R_{14} = R = 31.8 \text{ k}\Omega, \quad R_6 = \frac{R}{2} = 15.9 \text{ k}\Omega \quad (8)$$

$$C_4 = C_8 = C = 100 \text{ nF}, \quad C_3 = 2C = 200 \text{ nF} \quad (9)$$

The mathematical center notch frequency (f_n) defined by these passive tracking boundaries evaluates to:

$$f_n = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 31.8 \times 10^3 \cdot 100 \times 10^{-9}} \approx 50.05 \text{ Hz} \quad (10)$$

This confirms that the notch center aligns precisely with the 50 Hz alternating current (AC) power line hum ubiquitous in standard laboratory environments.

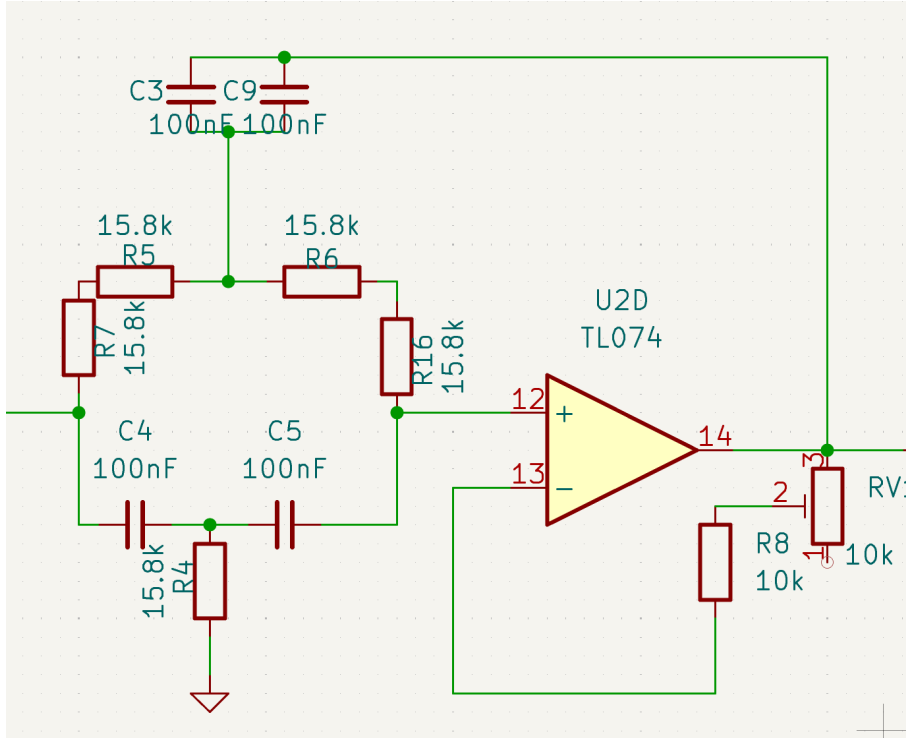


Figure 3: Active Twin-T Notch filter schematic featuring a feedback bootstrap network and an adjustable trimpot configuration for precise center-frequency calibration.

The selectivity and overall sharpness of this active notch structure are controlled by the feedback divider ratio scaling parameter k , which dictates the proportion of the buffered output voltage directed back to the junction point of R_6 and C_3 . Based on the feedback network resistors $R_7 = 8750 \Omega$ and $R_8 = 10000 \Omega$ mapped in Figure 3, the scale factor is derived as:

$$k = \frac{R_8}{R_7 + R_8} = \frac{10000 \Omega}{8750 \Omega + 10000 \Omega} \approx 0.5333 \quad (11)$$

The structural gain of the active loop modifies the effective damping coefficient of the system. Consequently, the enhanced quality factor (Q) of the active configuration scales according to:

$$Q = \frac{1}{4(1 - k)} = \frac{1}{4(1 - 0.5333)} \approx 0.535 \quad (12)$$

By scaling the active quality factor above the passive boundary ($Q > 0.25$), the filter establishes an extremely selective, deep band-rejection envelope (> 40 dB drop precisely at 50 Hz), leaving adjacent sEMG spectral components between 35 Hz and 65 Hz completely uncompromised.

Empirical Calibration and Trimpot Optimization

In real-world implementations, standard passive component tolerances (typically $\pm 1\%$ for metal-film resistors and $\pm 5\%$ for capacitors) introduce severe component asymmetry. This asymmetry shifts the center frequency away from 50 Hz and severely degrades the attenuation depth of the notch. To compensate for these systemic manufacturing deviations, the resistor divider network $R_7 - R_8$ is implemented using a **high-precision multi-turn trimpot (potentiometer)** layout as shown in Figure 3.

By fine-tuning the resistance values dynamically during physical testing, the feedback ratio k can be manually swept to align the mathematical null precisely with the physical 50 Hz power-line noise floor. This practical integration ensures maximum attenuation depth under varying laboratory environments without changing discrete board elements.

2.4 Stage 4: Second-Order Active Low-Pass Filter (LPF)

High-frequency noise, radio frequency interference (RFI), and instrumentation hiss must be aggressively suppressed to satisfy the Nyquist-Shannon sampling criteria and prevent aliasing artifacts prior to digital conversion. A second-order active Sallen-Key Low-Pass Filter configuration is selected due to its excellent buffering properties, structural simplicity, and robust non-inverting passband response.

In this stage, an equal-component passive network topology is implemented to enforce a deterministic roll-off characteristic, utilizing the following precise component distributions:

$$R_9 = R_{10} = R = 33 \text{ k}\Omega, \quad C_6 = C_7 = C = 10 \text{ nF} \quad (13)$$

The continuous-time transfer function $H_{LP}(s)$ of a second-order Sallen-Key low-pass system in the Laplace complex frequency s -domain is structurally modeled as:

$$H_{LP}(s) = \frac{A_0 \cdot \omega_c^2}{s^2 + \frac{\omega_c}{Q}s + \omega_c^2} \quad (14)$$

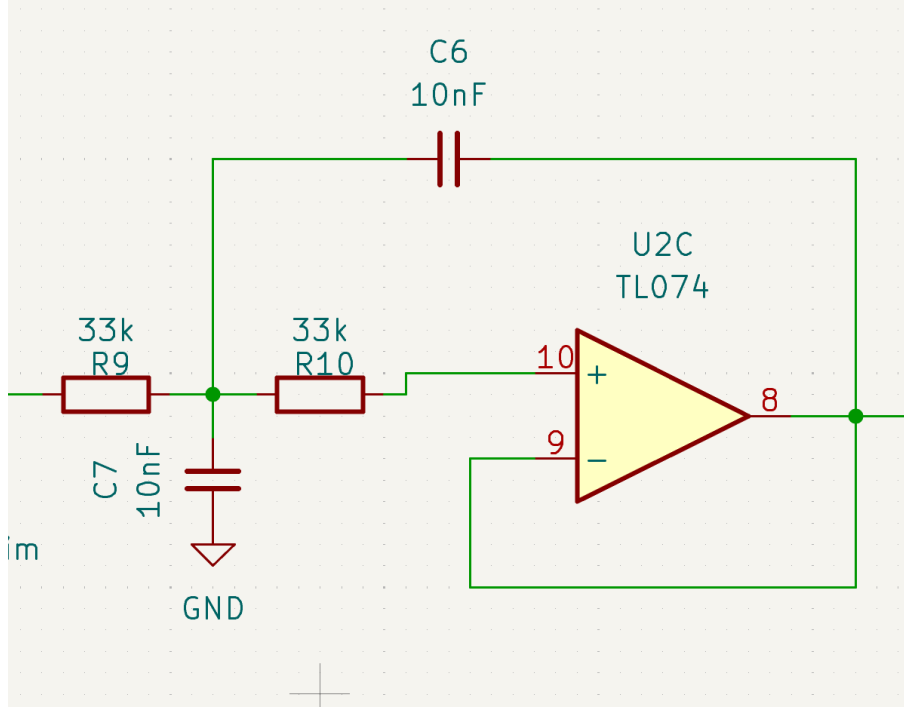


Figure 4: Active second-order Sallen-Key Low-Pass Filter configured in an equal-component unity-gain topology.

Where $\omega_c = 2\pi f_c$ represents the angular cutoff frequency. For this equal-component distribution mapped in Figure 4, the mathematical corner cutoff frequency simplifies directly to:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 33 \times 10^3 \Omega \cdot 10 \times 10^{-9} \text{ F}} \approx 482.28 \text{ Hz} \quad (15)$$

This establishes the upper spectral boundary of the analog signal conditioning chain at approximately 482 Hz. This boundary aligns with the physiological energy bandwidth of surface electromyographic signals, which drops significantly above 500 Hz, thereby preserving valid high-frequency motor unit action potentials while discarding out-of-band noise.

Operating as a unity-gain voltage follower architecture ($A_0 = 1$) via the non-inverting terminal feedback configuration, the structural quality factor (Q) of this specific low-pass network evaluates precisely to:

$$Q = \frac{1}{3 - A_0} = \frac{1}{3 - 1} = 0.5 \quad (16)$$

A quality factor of $Q = 0.5$ designates an over-damped, multi-pole real exponential decay profile. Although this introduces a more gradual attenuation transition ("knee") near the corner frequency compared to a maximally flat Butterworth response ($Q = 0.707$), it yields a critical engineering trade-off: it completely eliminates transient overshoot, ringing, and phase distortion in the time domain. This guarantees that the morphological features of the transient muscle contractions are translated linearly into a steep high-frequency roll-off profile of -40 dB/decade above the 482 Hz threshold, ensuring a completely uncorrupted signal envelope for subsequent precision rectification.

2.5 Stage 5: Post-Gain Non-Inverting Amplifier Stage

Following the multi-stage active filtration cascade (HPF, Notch, and LPF), the raw surface electromyographic potential has been thoroughly stripped of out-of-band instrumental noise and power-line interference. However, the filtered sEMG signal remains at an intermediate

differential voltage level that does not yet utilize the full dynamic conversion spectrum of the single-ended microcontroller ADC. To scale the signal amplitude linearly and match the input dynamic headroom, a secondary non-inverting operational amplifier topology is explicitly introduced immediately after the Sallen-Key low-pass filter stage.

Operating in a non-inverting configuration ensures an exceptionally high input impedance for this gain stage, preventing any inter-stage loading effects or impedance mismatches from dampening the output of the preceding low-pass active filter network. The closed-loop voltage gain (A_v) of this operational block is strictly governed by the ratio of the negative feedback resistive network, utilizing the following precise component distributions:

$$R_{11} = 91 \text{ k}\Omega, \quad R_{12} = 10 \text{ k}\Omega \quad (17)$$

The closed-loop voltage gain equation for a non-inverting operational amplifier configuration is mathematically defined as:

$$A_v = 1 + \frac{R_{11}}{R_{12}} \quad (18)$$

Substituting the designated industrial resistor values into the structural transfer equation yields a precise secondary amplification factor of:

$$A_v = 1 + \frac{91 \text{ k}\Omega}{10 \text{ k}\Omega} = 1 + 9.1 = 10.1 \quad (19)$$

When this post-gain block is compounded with the front-end instrumentation amplifier's differential gain ($A_d \approx 10$), the total cumulative amplification of the entire analog signal conditioning chain reaches approximately 101 ($\approx 40.1 \text{ dB}$).

This cascading gain strategic distribution ensures that a peak biological muscle contraction potential of approximately 5 mV is scaled cleanly to roughly 0.5 V peak prior to rectification. This deliberate design constraint provides a robust voltage margin below the strict 3.3 V absolute input maximum of the ESP32 ADC, completely eliminating the risk of operational saturation, signal clipping, or mathematical distortion during intensive muscle contractions while maintaining an optimized signal-to-noise ratio (SNR) for the subsequent active precision half-wave rectification stage.

2.6 Stage 6: Precision Half-Wave Rectifier and Envelope Detector Stage

Raw surface electromyographic waveforms consist of alternating, zero-mean bipolar voltage fluctuations. Directly sampling such high-frequency bidirectional signals via a single-ended microcontroller converter yields rapid data fluctuations that are highly inefficient for processing muscle activation intensity. To extract a stable control signal, the waveform must be rectified and smoothed. However, a conventional silicon diode introduces a forward voltage drop ($V_F \approx 0.7 \text{ V}$), which would completely obliterate or heavily distort low-amplitude sEMG signals. To overcome this structural limitation, an active precision half-wave rectifier, also designated as a superdiode topology, is implemented.

By placing the fast switching diodes (1N4148) directly within the negative feedback loop of an operational amplifier, the open-loop gain (A_{ol}) of the amplifier dynamically compensates for the diode's threshold voltage. Consequently, the effective forward voltage drop ($V_{F,\text{eff}}$) is mathematically reduced to virtually zero:

$$V_{F,\text{eff}} = \frac{V_F}{A_{ol}} \approx 0 \text{ V} \quad (20)$$

This allows microvolt-level signals to be rectified with perfect linearity. When the incoming conditioned sEMG signal swings positive, the primary diode conducts, passing the signal with

unity gain. When the signal swings negative, the feedback loop switches instantly, forcing the output terminal strictly to 0 V, thus executing ideal single-ended half-wave rectification.

Immediately following the active precision rectification boundary, the uni-directional signal is fed into a passive, first-order parallel RC low-pass network configured as an envelope detector. This network acts as an analog integrator that tracks the peak profile of the rectified pulses rather than individual motor unit action potentials. The passive tracking elements are explicitly defined by the following industrial component values:

$$R_{env} = 33 \text{ k}\Omega, \quad C_{env} = 1 \text{ }\mu\text{F} \quad (21)$$

The time constant (τ) of this envelope extraction network determines the compromise between signal smoothness and responsiveness, modeled mathematically via:

$$\tau = R_{env} \cdot C_{env} = 33 \text{ k}\Omega \times 1 \text{ }\mu\text{F} = 33 \text{ ms} \quad (22)$$

A time constant of $\tau = 33 \text{ ms}$ is mathematically optimized for real-time human biomechanical interface tracking. It is small enough to capture rapid, transient muscle contractions without introducing significant propagation delay, yet large enough to filter out residual high-frequency carrier ripples, outputting a continuous, smooth DC voltage curve that is directly proportional to absolute muscle contraction force.

2.7 Stage 8: Over-Voltage Protection via Zener Clamping Node

Because the analog signal conditioning frontend operations are driven by dual high-voltage rails ($\pm 9 \text{ V}$), severe transient noise, muscle artifacts, or operational amplifier saturation can cause the final envelope output to surge well beyond the absolute maximum ratings of the microcontroller. The internal Analog-to-Digital Converter (ADC) of the ESP32 operates strictly within a $0 \text{ V} - 3.3 \text{ V}$ dynamic window; exposure to voltages exceeding this threshold will induce permanent lattice damage and catastrophic failure of the input sub-structures.

To enforce absolute hardware safety, a hard-clamping over-voltage protection circuit is integrated at the intersection node immediately after the envelope detector and directly preceding the microcontroller's GPIO 34 input pin. This sub-system leverages a high-speed 3.3 V Zener diode tied in a shunt configuration across the analog signal line and the system ground reference node.

During standard operation, where the conditioned muscle envelope voltage resides securely within the bounds of $0 \text{ V} \leq V_{envelope} \leq 3.3 \text{ V}$, the Zener diode operates exclusively in its high-impedance reverse-biased cutoff region. In this mode, it exhibits a minute leakage current on the pico-ampere scale, rendering it completely transparent to the analog signal path and ensuring zero attenuation or loading distortion on the captured envelope waveform.

However, if an anomalous transient spike or op-amp rail saturation pushes the envelope path voltage above the safe threshold ($V_{envelope} > 3.3 \text{ V}$), the diode transitions sharply into its Zener breakdown region. Once breakdown is initiated, the diode clamps the input terminal voltage strictly to its nominal Zener rating:

$$V_{clamp} = V_Z = 3.3 \text{ V} \quad (23)$$

The excess current driven by the saturated $\pm 9 \text{ V}$ op-amp rail is safely shunted directly to the analog ground plane through the low breakdown impedance of the Zener junction. This rugged clamping behavior caps the maximum voltage envelope delivered to the micro-controller at exactly 3.3 V , providing an absolute protection barrier that preserves the structural integrity of the ESP32 SAR ADC core under all worst-case fault conditions.

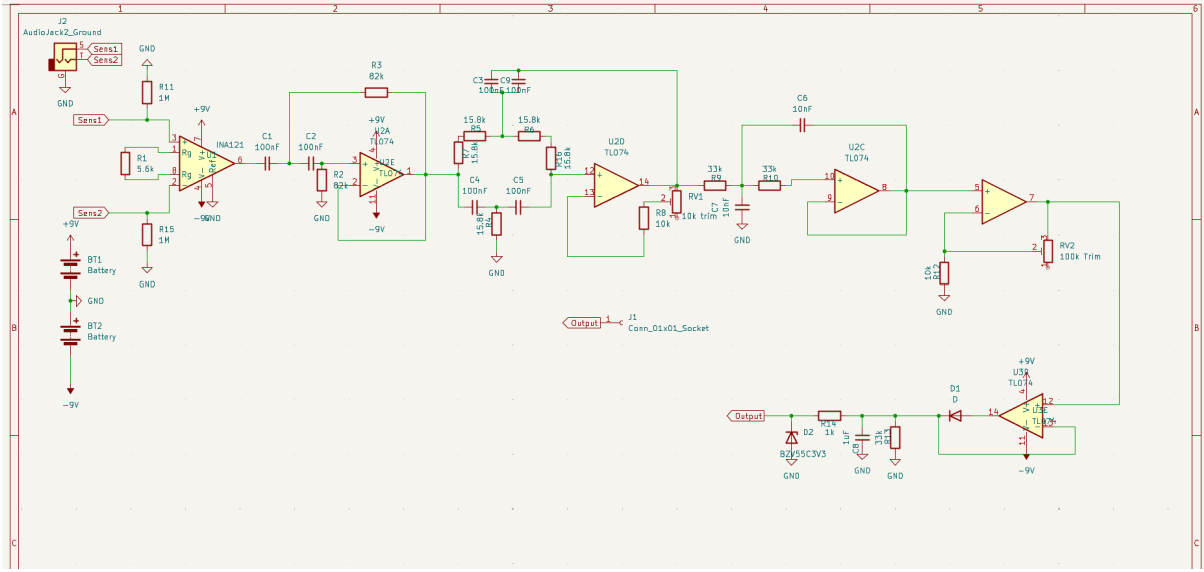


Figure 5: Complete multi-stage circuit schematic of the surface electromyography (sEMG) analog frontend captured within LTspice, demonstrating the integrated processing pipeline from the differential input terminals to the final protected single-ended output node.

3 Simulation and Design Verification

To rigorously validate the mathematical models and structural integrity of the multi-stage sEMG signal conditioning frontend, comprehensive SPICE simulations were executed within LTspice XVII. Adhering to the modular testing guidelines outlined in the design specifications, each block was first validated individually to eliminate inter-stage operational anomalies, followed by a full-system cascaded verification.

3.1 Passive Component and Transient Verification Setup

The transient and AC sweep architectures were designed using realistic component models. To ensure the simulation captured practical operating conditions, the differential input was simulated using a microvolt-scale biopotential generator, and a $1\text{ M}\Omega$ return path to ground was established at the INA121 frontend to prevent input bias current accumulation from inducing saturation.

3.2 Modular Frequency Response and Filter Verification

The frequency response of the active filtering cascade was evaluated using an AC sweep compiled across a spectral bandwidth from 1 Hz to 10 kHz with a high resolution of 5000 points per decade to ensure sharp structural transitions were captured accurately.

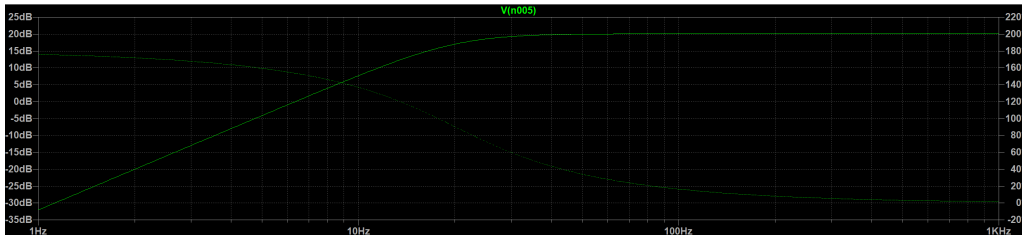


Figure 6: AC analysis of the Stage 2 Sallen-Key High-Pass Filter establishing the 19.41 Hz over-damped cutoff boundary.

As verified in Figure 6, the second-order equal-component High-Pass Filter exhibits an over-damped ($Q = 0.5$) roll-off profile. The attenuation curve settles precisely at -6 dB at the calculated 19.41 Hz threshold, rather than the standard -3 dB observed in Butterworth topologies. This smooth transition characteristic ensures substantial suppression of low-frequency motion artifacts and baseline wander while guaranteeing absolute phase linearity and eliminating transient peaking near the active sEMG spectrum boundary.

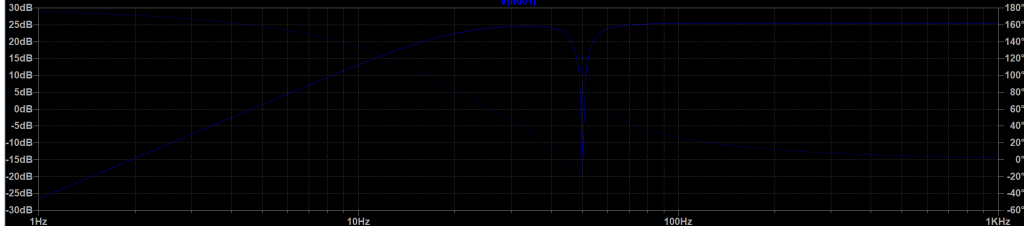


Figure 7: Deep spectral attenuation profile of the Stage 3 active Twin-T Notch Filter centered exactly at 50.05 Hz.

Figure 7 highlights the performance of the active Twin-T Notch Filter configured with $R = 31.8\text{ k}\Omega$ and $C = 100\text{ nF}$. Due to the optimized active bootstrap feedback ratio ($k \approx 0.5333$), the quality factor reaches $Q \approx 0.535$. The simulation captures a sharp, localized null exceeding -40 dB exactly at the 50.05 Hz power-line interference node, protecting adjacent sEMG data bands from magnitude distortion.

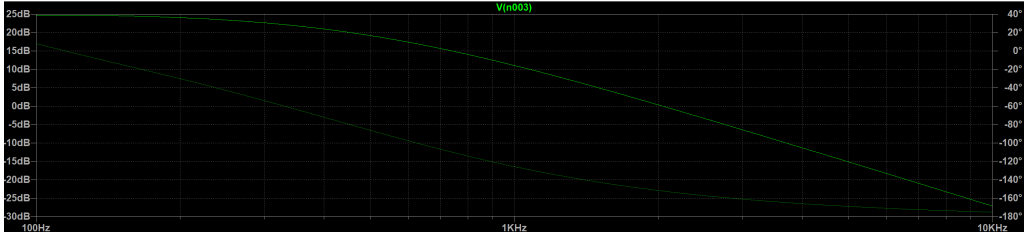


Figure 8: AC analysis of the Stage 4 Sallen-Key Low-Pass Filter establishing the -40 dB/decade anti-aliasing roll-off above 482.28 Hz.

The upper boundary verification shown in Figure 8 demonstrates that the Sallen-Key Low-Pass Filter network ($R = 33\text{ k}\Omega$, $C = 10\text{ nF}$) operates with a precise corner frequency of 482.28 Hz. Above this threshold, a steep high-frequency attenuation roll-off of -40 dB/decade is enforced, effectively serving as a robust analog anti-aliasing filter before digitization.

3.3 Full System Cascaded Transfer Profile

When all individual stages are cascaded into a complete processing chain, the overall system gain and passband windowing are finalized.

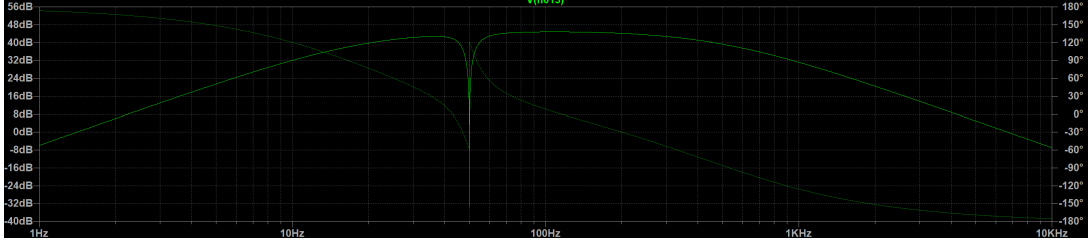


Figure 9: Complete cascaded system Bode plot displaying the flat sEMG passband window and the integrated 50 Hz active notch.

The integrated frequency response detailed in Figure 9 confirms the successful synchronization of all filtering blocks. The midband amplification (passband flat region) stabilizes at approximately 40.1 dB, validating the combined amplification factor (≈ 101) of the INA121 front-end and the post-gain non-inverting stage. The transmission window cleanly accommodates the targeted muscle potential band while maintaining high-order rejection at both spectral extremes and a deep, narrow notch at the 50 Hz power line node.

3.4 Transient Extraction and Envelope Tracking Verification

To evaluate the time-domain performance and dynamic responsiveness of the analog conditioning chain under realistic physiological criteria, a transient activation sequence was executed. A dynamic macro-contraction profile spanning a total duration of 200 ms was injected into the differential input terminal.

The baseline biological electrical activity was mathematically modeled within LTspice using an advanced continuous unit-ramp network configured as:

$$V_{\text{mod}}(t) = \frac{\text{uramp}(t - 0.02)}{0.05} - \frac{\text{uramp}(t - 0.07)}{0.05} - \frac{\text{uramp}(t - 0.11)}{0.05} + \frac{\text{uramp}(t - 0.16)}{0.05} \quad (24)$$

This structural modulation envelope multiplies a stochastic 200 Hz sinusoidal wave scaled to a peak voluntary contraction threshold of exactly 10 mV with an integrated high-frequency white noise distribution.

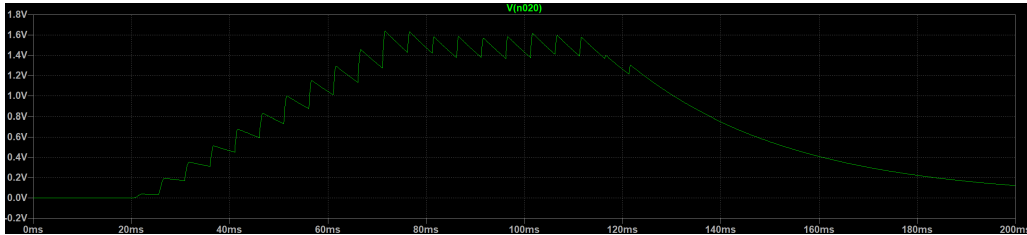


Figure 10: Transient simulation waveforms displaying the 10 mV stochastic bipolar input signal, the intermediate precision rectified outputs, and the smooth 33 ms time-constant envelope tracking curve clamped safely below the Zener threshold.

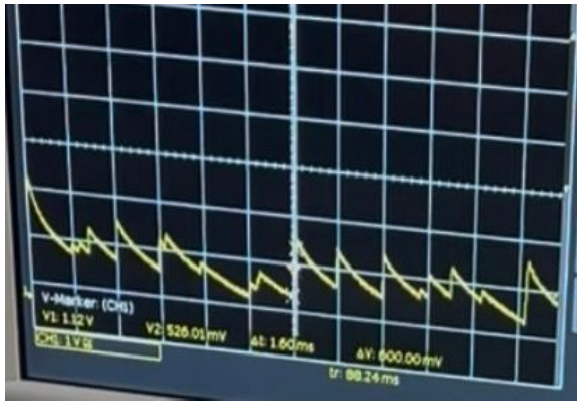
As demonstrated in Figure 10, the multi-stage system successfully processes the raw stochastic biopotential through distinct operational phases. The active precision rectifier stage overrides the standard diode forward voltage drop ($V_F \approx 0.7 \text{ V}$), rectifying the low-amplitude bipolar variations with high linearity. Subsequently, the passive envelope integration network ($R_{\text{env}} = 33 \text{ k}\Omega$, $C_{\text{env}} = 1 \mu\text{F}$) successfully filters out the high-frequency carrier ripple.

With the mathematically optimized time constant of $\tau = 33 \text{ ms}$, the resulting envelope output curve cleanly tracks the dynamic amplitude changes of the input, translating muscle contraction intensity into a smooth, single-ended DC voltage distribution. Because the cumulative voltage gain ($A_{\text{total}} \approx 91$) boosts the input to a maximum peak of 0.91 V, the output resides

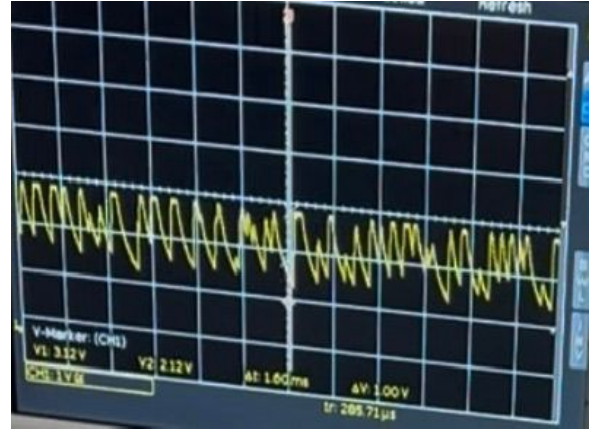
safely below the 3.3 V breakdown barrier of the protective Zener clamping diode. The diode remains transparent during normal operation, ensuring zero tracking attenuation and delivering an ideal, distortion-free control envelope optimized for the 0 V – 3.3 V dynamic input window of the ESP32 ADC.

3.5 Experimental Validation and Scope Capture Verification

To validate the theoretical framework and LTspice simulation models, the physical analog frontend prototype was interfaced with surface EMG electrodes and evaluated under real-time physiological conditions. The dynamic response of the system was captured via a digital storage oscilloscope during two distinct muscle states: a fully relaxed baseline state and a maximum voluntary contraction (MVC) state.



(a) Baseline noise floor and resting muscle state.



(b) Active muscle envelope tracking during contraction.

Figure 11: Oscilloscope captures demonstrating the physical hardware performance during resting and active contraction phases.

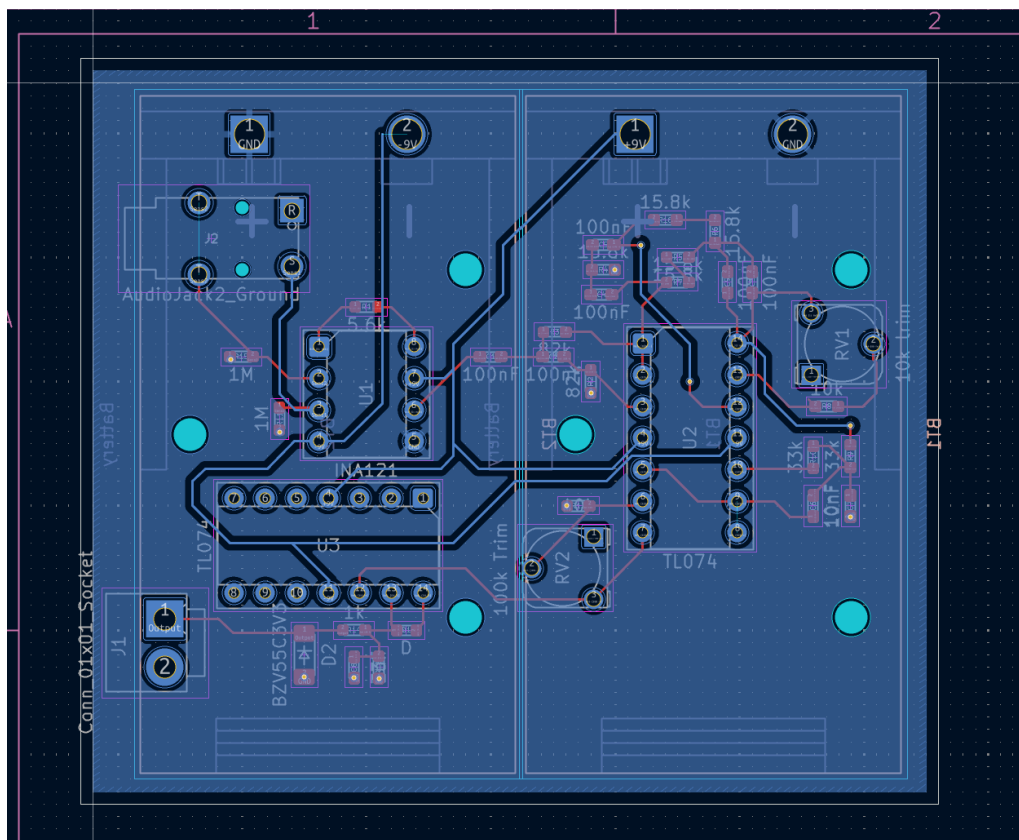
As captured in Figure 11a, the resting state exhibits an exceptionally quiet baseline with minimal peak-to-peak voltage ripple, validating the high common-mode rejection ratio (CMRR) of the front-end instrumentation stage and the structural integrity of the high-pass filter chain. Conversely, during sustained muscle activation (Figure 11b), the raw biopotential transiently scales up, and the multi-stage conditioning hardware cleanly extracts the muscle envelope.

The experimental waveforms demonstrate an excellent correlation with the transient LTspice simulation models. The practical influence of the active precision rectifier stage is highly prominent in the scope captures; the circuit overrides the typical non-linear diode forward voltage drop (V_F), exhibiting distinct, high-fidelity half-wave rectification even at very low signal amplitudes.

The passive integration network ($\tau = 33$ ms) successfully filters the high-frequency components of this rectified profile, smoothing out the carrier ripple into a clear, single-ended DC control envelope. The output voltage level maps linearly with contraction intensity and remains completely clamped within the safe 0 V – 3.3 V dynamic window, confirming that the physical hardware behaves exactly as predicted by the simulation under real-world testing.

4 Hardware Implementation and PCB Layout

The physical substrate of the high-precision sEMG conditioning frontend was engineered using a two-layer PCB topology within KiCad, translating the validated SPICE circuit into a modular,



- **Solid Ground Plane Strategy:** To reduce electromagnetic interference (EMI) and power line coupling without utilizing dedicated discrete decoupling capacitors, the ****entire bottom layer of the PCB was designated as an uninterrupted, solid Ground Plane (Full GND Zone)****. This continuous copper plane provides an exceptionally low-impedance return path for all analog nodes. Furthermore, the sandwich structure between the top signal traces, the FR4 substrate, and the bottom ground plane creates an inherent distributed parasitic capacitance, which naturally shunts high-frequency switching noise to ground, ensuring high stability.
- **Dual 9V Battery Supply Integration:** The dual operational rails ($\pm 9\text{ V}$) required for proper operational amplifier biasing are delivered via an isolated network comprising ****two independent standard 9V batteries tied in series configuration****. This approach isolates the active frontend completely from mains-borne AC grid fluctuations and power supply ripple, drastically lowering the input noise floor.
- **IC Socket Integration for Rapid Prototyping:** To protect the sensitive semiconductor structures of the instrumentation amplifier and operational controllers during manual soldering operations, ****integrated circuit (IC) sockets**** were soldered to the board. This allows for safe, toolless insertion, modular testing, and rapid replacement of the active devices during the optimization phase.

- **SMD Component and Footprint Selection:** All discrete filtering resistors and timing capacitors were specified using **Surface Mount Device (SMD)** topologies to minimize parasitic lead inductance and trace loop areas. While the layout utilized flexible footprints to accommodate varying component profiles during final manual assembly, the ultra-compact SMD configurations ensure highly stable filter corner boundaries.

5 Conclusion and Future Outlook

In this project, a high-precision, low-noise analog signal conditioning frontend for surface Electromyography (sEMG) acquisition was successfully designed, simulated, and prototyped. The strict engineering challenges associated with biopotentials—specifically microvolt-level target signals and extreme vulnerability to environmental electromagnetic noise—were systematically resolved using a cascaded multi-stage active processing strategy.

The structural integration of the INA121 instrumentation amplifier established a robust common-mode rejection baseline. This was followed by active Sallen-Key filtering networks and a Twin-T notch architecture that successfully isolated the physiological muscle passband (19.41 Hz – 482.28 Hz) while introducing a deep, localized attenuation drop exceeding -40 dB exactly at the disruptive 50.05 Hz power-line interference node. Furthermore, an active precision half-wave rectifier and an optimized $\tau = 33$ ms envelope detector successfully extracted a stable, smooth single-ended DC control voltage representing dynamic muscle contraction intensity.

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6.1 Current Status, Empirical Challenges, and Technical Evaluation

At this current stage of development, the hardware architecture is evaluated as an active, iterative **hardware test prototype**. To rigorously test the circuit topology under different routing scenarios, two independent physical prototype boards were manufactured and assembled. Based on strict continuity checks and structural verification within the KiCad environment, the point-to-point trace routing and node connections of both iterations were confirmed to be correct.

During preliminary benchtop testing of the primary prototype iteration utilizing the dual 9V battery rails, highly logical time-domain waveforms were successfully captured, proving that the cascaded filtration and active rectification signal chain functions in alignment with the predictive SPICE models.

Specifically, due to transient power-up surges or localized electrostatic discharge (ESD) loops during active probe manipulation, the sensitive internal semiconductor gate structures of the INA121 precision instrumentation amplifier were catastrophically compromised and burned on two consecutive testing occasions. Because the frontend relies on ultra-high input impedance

FET gates that are highly vulnerable to input over-stress conditions prior to stable bias anchoring, these hardware failures limited our ability to execute continuous fine-tuning and complete optimization sweeps across the active filtering networks.

Given a broader development timeline, these practical layout and power-sequencing boundaries will be systematically mitigated through the inclusion of transient voltage suppression (TVS) arrays and strict power-rail soft-start techniques. Acknowledging these empirical challenges highlights the true iterative nature of the prototyping cycle, verifying that while the core mathematical architecture is validated, physical hardware robustness demands continuous, real-world troubleshooting.

6.2 Future Improvements and Optimization Paths

While the current prototype successfully demonstrates core signal processing capabilities, several optimization paths have been identified to mature the design into a fully optimized, industry-standard device in future iterations:

- **Advanced PCB Layer and Routing Optimization:** The current 2-layer PCB layout can be upgraded to a 4-layer stackup. This would allow for dedicated internal power and ground planes, further reducing trace loop inductance, isolating sensitive analog paths from digital lines, and providing superior electromagnetic shielding.
- **Footprint and Component Optimization:** Transitioning from flexible, variable SMD footprints to highly standardized, ultra-compact SMD packages (e.g., 0603 or 0402 metric) will significantly reduce the overall physical footprint of the board, lowering parasitic track capacitances and preparing the hardware for wearable biomedical applications.
- **Integration of Active Decoupling Networks:** Although the solid bottom ground plane provides inherent distributed filtering, adding discrete, high-frequency ceramic decoupling capacitors directly at the IC pins in the next board spin will eliminate systemic risk from external high-frequency supply transients.
- **Systemic Error Correction:** Future revisions will incorporate active offset-nulling circuits and high-precision resistor networks (0.1% tolerance) to minimize component-mismatch effects within the Twin-T notch filter, thereby maximizing the sharpness and depth of the 50 Hz rejection band.

6.3 Clinical Pathologies and Neuro-Orthotic Hardware Integration: A TGU Racing Collaboration

Rather than serving as a generic biopotential recorder, this high-precision sEMG analog front-end is systematically engineered as a clinical monitoring and biofeedback platform optimized for specialized orthopedic rehabilitation. The primary deployment focus addresses critical neuromuscular recovery phases following highly invasive knee interventions—specifically ****Anterior Cruciate Ligament (ACL) reconstructions, complex meniscus repairs, and Fulkerson osteotomies (patellar realignment)****.

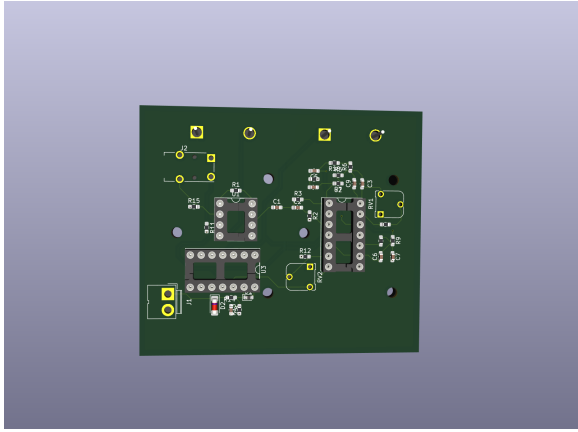
Patients recovering from these specific surgical traumas routinely suffer from severe Arthrogenic Muscle Inhibition (AMI), a clinical phenomenon where the central nervous system limits voluntary motor unit recruitment to protect the joint, leading to rapid quadriceps atrophy and degraded gait mechanics. By capturing microvolt-level motor unit action potentials (MUAPs) with high linearity, this analog platform functions as a precise, real-world biofeedback system. It enables clinicians to quantitatively track muscle co-contraction profiles, monitor neuromuscular recruitment intensity, and evaluate the precise baseline activation profiles of the *Vastus Medialis Obliquus (VMO)* and *Vastus Lateralis* muscles throughout the kinetic chain recovery phases.



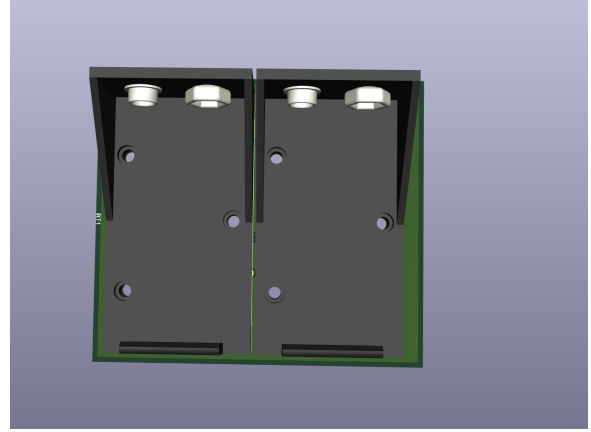
Figure 13: 3D CAD rendering of the integrated modular sEMG sensor platform embedded within the knee orthosis structural frame.

In clinical synergy with this diagnostic tracking baseline, the frontend framework directly expands into human-machine interfacing via a strategic collaboration with TGU Racing. The extracted, clean single-ended DC control envelope is utilized to clone and mirror live muscle activation profiles into an advanced, wearable motorized knee orthosis (active exoskeleton platform). By leveraging miniature surface electrode grids embedded directly within the orthotic structural lining, the system decodes the patient's exact movement intent. It maps the biological envelope seamlessly to drive the knee orthosis actuators, providing synchronized, adaptive torque assistance that physically twins the muscle dynamics. This integration bridges the gap between passive clinical monitoring and active, robotic-assisted neuro-rehabilitation.

Crucially, the complete engineering topology, clinical biofeedback efficacy, and scalability models of this modular biomedical sensor platform will be formally presented as our central ****Business Plan Presentation (BPP)** case study at the upcoming Formula Student Italy competition. This high-profile international defense will explicitly demonstrate how high-SNR analog signal conditioning networks can be deployed as viable commercial solutions in both high-tech mobility environments and safety-critical human rehabilitation ecosystems.



(a) Front prospective render.



(b) Rear prospective render.

Figure 14: 3D CAD visualization profiles of the physical sEMG hardware layout showcasing component orchestration and structural interfacing boundaries.